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CPCS-214 Computer Architecture and Organization

43 - Cache Memory

Introduction

- *Cache* was the name chosen to represent the level of the memory hierarchy between the processor and main memory in the first commercial computer to have this extra level
- Memories in datapath are simply replaced by caches

Cache

- We begin by looking at a very simple cache
- Processor requests are each one word
- Blocks also consist of a single word
- Figure shows such a cache, before and after requesting a data item that is not initially in the cache

Reference Request

- Before request, cache contains a collection of recent references $X_1, X_2, \dots, X_{n-1}$
- Processor requests a word X_n that is not in cache
 - request results in a miss
 - word X_n is brought from memory into cache

Reference Request

X_4
X_1
X_{n-2}
X_{n-1}
X_2
X_3

a. Before the reference to X_n

Reference Request

X_4
X_1
X_{n-2}
X_{n-1}
X_2
X_3

a. Before the reference to X_n

X_4
X_1
X_{n-2}
X_{n-1}
X_2
X_n
X_3

b. After the reference to X_n

Direct Mapped Cache

- How do we know if a data item is in the cache?
- Moreover, if it is, how do we find it?
- Answers are related
- If each word can go in exactly one place in the cache, then it is straightforward to find the word if it is in the cache

Direct Mapped Cache

- Simplest way to assign location in cache for each word in memory is to assign cache location based on *address* of word in memory
- This structure is called **direct mapped**
 - since each memory location is mapped directly to exactly one location in cache

Direct Mapped Cache

- Direct-mapped caches use this mapping to find a block

(Block address) modulo (Number of blocks in cache)

- If number of entries in cache is power of 2, then *modulo* can be computed simply by using
low-order $\log_2$ (cache size in blocks) bits of address
- Thus, 8-block cache uses 3 lowest bits ($8 = 2^3$) of block address

Direct Mapped Cache

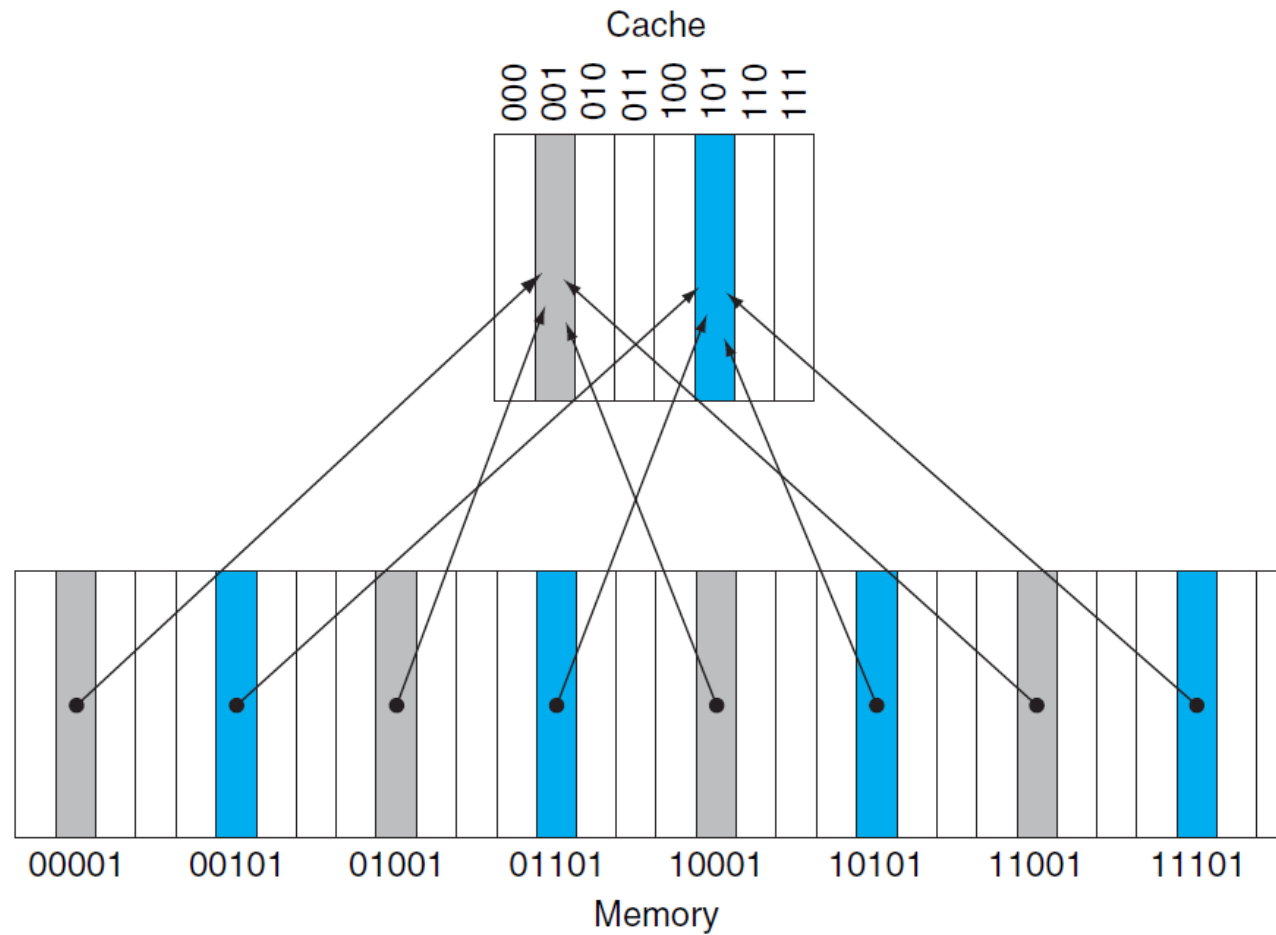
- For example, figure shows how memory addresses between

$1_{\text{ten}} = (00001_{\text{two}})$ and $29_{\text{ten}} = (11101_{\text{two}})$ map to locations

$1_{\text{ten}} = (001_{\text{two}})$ and $5_{\text{ten}} (101_{\text{two}})$

in a direct-mapped cache of 8 words

Direct Mapped Cache



Tag

- Each cache location can contain contents of a number of different memory locations
- How do we know whether data in cache corresponds to a requested word?
- How do we know whether a requested word is in cache or not?
- By adding a set of **tags** to the cache

Tag

- Tags contain address information required to identify whether a word in the cache corresponds to the requested word
- Tag needs only to contain the upper portion of the address, corresponding to bits that are not used as an index into cache

Tag

- For example, in the previous figure we need only have upper 2 of 5 address bits in tag, since lower 3-bit index field of address selects block
- Architects omit the index bits because they are redundant, since by definition index field of any address of cache block must be that block number

Valid Bit

- A way to recognize that a cache block does not have valid information is needed
 - For instance, when processor starts up, cache does not have good data, and tag fields will be meaningless
 - Even after executing many instructions, some of cache entries may still be empty
 - Thus, need to know that tag should be ignored for such entries

Valid Bit

- Most common method is to add a **valid bit** to indicate whether an entry contains a valid address
- If bit is not set, there cannot be a match for this block

Accessing a Cache

- An empty eight-block cache
- Since there are 8 blocks in cache, low-order 3 bits of address give block number

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

a. The initial state of the cache after power-on

Accessing a Cache

- Sequence of nine memory references are shown next
- Next figures show how contents of cache change on each memory reference
- Since there are 8 blocks in cache, low-order 3 bits of address give block number

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110 _{two}		
26	11010 _{two}		
22	10110 _{two}		
26	11010 _{two}		
16	10000 _{two}		
3	00011 _{two}		
16	10000 _{two}		
18	10010 _{two}		
16	10000 _{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}		$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		Memory (10110_{two})
111	N		

Handling a miss of address (10110_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (10110_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (10110_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}		
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (11010_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10\mathbf{110}_{\text{two}} \bmod 8) = \mathbf{110}_{\text{two}}$
26	11010_{two}	miss	$(11\mathbf{010}_{\text{two}} \bmod 8) = \mathbf{010}_{\text{two}}$
22	10110_{two}		$(10\mathbf{110}_{\text{two}} \bmod 8) = \mathbf{110}_{\text{two}}$
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}		$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}		
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (10000_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}		$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}		
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (00011_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}		$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}	hit	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}		
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}	hit	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}		$(10010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}	hit	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}	miss	$(10010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}	hit	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}	miss	$(10010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	10_{two}	Memory (10010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Handling a miss of address (10010_{two})

Accessing a Cache

Decimal address of reference	Binary address of reference	Hit or miss in cache	Assigned cache block (where found or placed)
22	10110_{two}	miss	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	miss	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
22	10110_{two}	hit	$(10110_{\text{two}} \bmod 8) = 110_{\text{two}}$
26	11010_{two}	hit	$(11010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}	miss	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
3	00011_{two}	miss	$(00011_{\text{two}} \bmod 8) = 011_{\text{two}}$
16	10000_{two}	hit	$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$
18	10010_{two}	miss	$(10010_{\text{two}} \bmod 8) = 010_{\text{two}}$
16	10000_{two}		$(10000_{\text{two}} \bmod 8) = 000_{\text{two}}$

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	10_{two}	Memory (10010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

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Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	10_{two}	Memory (10010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

Referenced Address Subdivision

- A referenced address is divided into
 - *Tag field*, used to compare with value of tag field of cache
 - *Cache index*, used to select block
- **Index** with **tag**, uniquely specifies memory address of word contained in cache block
- Since words are aligned to multiples of 4 bytes, least significant 2 bits of every address specify a byte within a word
 - Hence, least significant 2 bits are ignored when selecting a word in the block

Referenced Address Subdivision

